Studying the off-target receptors of common drugs

A PROJECT REPORT

Submitted by:

Bhavika Bisht

(245HSBB008)

to

Institute of Bioinformatics and Applied Biotechnology

in partial fulfilment of the requirements for

Master of Science in Biotechnology and Bioinformatics

(Degree to be awarded by Bangalore University, Bengaluru)

under the guidance of:

Prof. S. Thiyagrajan

IBAB

Institute of Bioinformatics and Applied Biotechnology

INSTITUTE OF BIOINFORMATICS AND APPLIED BIOTECHNOLOGY

BENGALURU

May 2026

DECLARATION

I certify that

- The work contained in this report is original and has been done by me under the guidance of my supervisor(s).

- I have followed the guidelines provided by the Department in preparing the report.

- I have conformed to the norms and guidelines given in the Honor Code of Conduct of the Institute.

- Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Signed by:

Bhavika Bisht

CERTIFICATE

It is certified that the work contained in this report titled ***"Studying the off-target receptors of common drugs"*** is the original work done by **Bhavika Bisht** and has been carried out under my supervision.

Signed by:

Prof. S. Thiyagrajan

Date: 29/5/26

INSTITUTE OF BIOINFORMATICS AND APPLIED BIOTECHNOLOGY
BENGALURU

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Prof. S. Thiyagarajan for his constant guidance, encouragement, and invaluable scientific insights throughout the course of this dissertation work. His constructive feedback, critical discussions, and continuous support played a major role in shaping the direction and quality of this study. I am especially thankful for the opportunity to work under his guidance and for the intellectually stimulating research environment that encouraged independent thinking and scientific curiosity.

I also extend my sincere appreciation to all faculty members, research scholars, and staff of the Institute of Bioinformatics and Applied Biotechnology for providing a supportive academic environment and access to the computational resources necessary for carrying out this work. Their discussions, suggestions, and technical support contributed significantly to my learning experience throughout this project.

I would also like to acknowledge the researchers and public repositories whose openly available datasets and published studies formed the foundation of this work. Their contribution to open scientific data sharing made this comparative transcriptomic analysis possible.

I am deeply grateful to my parents and family for their constant encouragement, patience, and unwavering support throughout my academic journey. Their motivation and belief in me have always been a source of strength. I also thank my friends, Pranjali and Advait, and colleagues for their encouragement, discussions, and moral support during the completion of this dissertation.

The first draft of certain sections of this report was generated with the assistance of AI-based language tools and was subsequently reviewed, refined, validated, and edited extensively by the author. All scientific analyses, biological interpretations, coding workflows, results, and conclusions presented in this dissertation were independently carried out and verified by the author.

Finally, I gratefully acknowledge the Institute of Bioinformatics and Applied Biotechnology (IBAB), Bengaluru, for providing the infrastructure and academic support necessary for this research work. This work was supported by the Department of Biotechnology – Bioinformatics Infrastructure Facility (DBT-BIF/BIC) and the Department of Electronics, Information Technology, Biotechnology and Science & Technology, Government of Karnataka.

ABSTRACT

Numerous small-molecule drugs engage with various protein targets, a phenomenon termed polypharmacology. Although these interactions may provide therapeutic advantages, they often result in Adverse Drug Reactions (ADRs). Diphenhydramine (DPH), a first-generation antihistamine, acts as a model multi-target drug recognized for its interaction with off-target receptors. Nevertheless, there is presently an absence of a comprehensive, proteome-wide interaction map for DPH. This research creates a high-throughput computational system to chart the off-target polypharmacological landscape of DPH. A detailed docking-ready dataset consisting of 25,820 distinct human protein structures was compiled, combining high-resolution experimental structures from the Protein Data Bank (PDB) with models generated by AlphaFold2. Employing a multi-layered molecular docking framework that includes DoGSiteScorer, the HYDE scoring method, and Autodock Vina, binding affinities across the proteome were measured. Functional enrichment and network topology analyses were then conducted utilizing STRING and KEGG databases. The findings effectively confirmed established off-target interactions with NMDA and Muscarinic receptors. Additionally, pathway analysis uncovered 19 significantly enriched novel biological pathways (FDR < 0.05), particularly highlighting critical metabolic weaknesses such as Fatty Acid Biosynthesis (45.3-fold enrichment; target: FASN) and Folate Biosynthesis (43.7-fold enrichment; targets: DHFR, FOLH1). Tissue-specific mapping conducted with the Human Protein Atlas identified these interactions within particular organ systems. This research showcases a highly scalable and reproducible method for proteome-wide off-target analysis, laying an essential groundwork for anticipating organ-specific toxicity and enhancing preclinical drug safety.

\

TABLE OF CONTENTS

List of Tables

List of figures

List of Abbreviation

Abbreviation	Description
ADR	Adverse Drug Reactions
AF2	AlphaFold2
CNS	Central Nervous System
DPH	Diphenhydramine
FDR	False Discovery Rate
GPCR	G-Protein Coupled Receptors
HPA	Human Protein Atlas
HTVS	High-Throughput Virtual Screening
KEGG	Kyoto Encyclopedia of Genes and Genomes
NMDA	N-Methyl-D-aspartate
OTC	Over The Counter
PDB	Protein Data Bank
PPI	Protein-Protein Interaction

QC	Quality Check
RCSB	Research Collaboratory for Structural Bioinformatics

CHAPTER 1
INTRODUCTION

1.1 Polypharmacology and Drug Safety

The traditional method in drug discovery relies on the principle of "one drug-one target," where a particular drug is developed to interact with a target protein to produce a specific outcome. Nevertheless, it is currently well recognized that many small-molecule drugs impact multiple targets, a concept known as polypharmacology, even if these are not always the intended targets (Rao et al., 2019; Sliwoski et al., 2014).

While such multiple target interactions are beneficial for managing complex illnesses, they are typically linked more often to adverse drug reactions (ADRs), side effects, and variable responses to specific medications (Rao et al., 2019). Off-target effects typically occur due to instances where several proteins possess similar binding pocket structures, amino acid components, or physicochemical properties. As a result, a medication designed for a specific protein is probably to interact or attach to other unrelated proteins too (Sliwoski et al., 2014).

Revealing these unintended interactions is helpful for enhancing drug safety and comprehending the mechanisms behind drug side effects. Nevertheless, experimentally linking these interactions through different methods has been shown to be both time-intensive and expensive, particularly with the involvement of thousands of proteins. This has created an important opportunity for structure-based molecular docking to serve as a method for large-scale prediction of drug-protein interactions (Rao et al., 2019).

The accessibility of resources from the protein data bank (PDB) (Berman et al., 2000; RCSB PDB, 2023), along with the growing number of resolved protein structures and new developments in AI-driven drug prediction tools such as AlphaFold (Jumper et al., 2021; Hayama et al., 2025), has enabled the execution of proteome-wide drug docking experiments. This enables the scientific community to efficiently evaluate drug compounds against numerous human targets cost-effectively and anticipate possible off-target effects of drugs in various human proteome tissues.

1.2 Diphenhydramine as a Model Compound

Diphenhydramine is a first-generation H1-antihistamine that has been commonly utilized for numerous decades (PubChem, 2023). It is among the most frequently sold over-the-counter medications, marketed under well-known names such as Benadryl, Nytol, and Unisom, and is utilized for managing allergic reactions, motion sickness, nausea, and brief sleep issues. Its mode of action involves the antagonism of the histamine H1 receptor, a G-protein-coupled receptor that plays a role in allergic reactions and the control of alertness (Hauser et al., 2017).

Unlike second-generation antihistamines, diphenhydramine is lipophilic and easily crosses the blood-brain barrier. This trait explains the notable CNS effects of the drug, such as sedation, drowsiness, and reduced cognitive alertness (PubChem, 2023). Although these CNS effects may be helpful in specific contexts, they also lead to various drug side effects.

Even with the availability of newer, more targeted antihistamines, diphenhydramine remains widely used due to its cost, availability as an over-the-counter option, and prolonged duration of effectiveness.

1.3 Structural Basis of H1 Receptor Binding and Promiscuity

The human H1 receptor (HRH1) is a Class A G-protein coupled receptor (GPCR) that promotes the pro-inflammatory effects of histamine. Structural analysis of the HRH1-doxepin complex (PDB ID: 3RZE) reveals a significant hydrophobic binding pocket located between the transmembrane helices (TMs). A key feature of this pocket is the conserved Aspartic Acid at position 107 (Asp107) in TM3 (Shimamura et al., 2011). This residue forms a crucial ionic "salt bridge" interaction with the protonated amine group of first-generation antihistamines, including Diphenhydramine (DPH).

The interaction between Aspartate and amine is a common motif in many biogenic amine receptors (including Dopamine, Serotonin, and Muscarinic receptors), thereby establishing the primary structural basis for the recognized polypharmacology of DPH. This study suggested that similar acidic residues present in non-target proteins might affect the off-target profile, necessitating a comprehensive structural and energetic assessment of the proteome.

1.4 Study Rationale

Understanding the off-target pharmacology of frequently used drugs is crucial for guaranteeing safety and minimizing adverse drug effects (Rao et al., 2019). Structural-based computational techniques are quickly becoming the solution for extensive in silico predictions of potential drug–protein interactions. The accuracy of these forecasts, however, relies on the quality of the protein structural dataset (Sliwoski et al., 2014).

The objective of this study was to assess and analyze the possibility of off-target interactions of the molecule diphenhydramine with different tissues of the human proteome through the technique of molecular docking. A dataset containing protein structures was created by combining PDB structures (Berman et al., 2000; RCSB PDB, 2023) with AI-derived AlphaFold structures (Jumper et al., 2021; Hayama et al., 2025).

A key emphasis was placed on tissue-specific protein expression, especially regarding proteins in the central nervous system, as diphenhydramine can successfully traverse the blood-brain barrier and displays substantial activity in the central nervous system.

Overall, this study intends to:

- Determine the possible secondary targets of diphenhydramine.
- Developing superior structural datasets appropriate for comprehensive docking studies.
- Demonstrate a reproducible computational method for off-target assessment throughout the whole proteome.

The findings obtained in this research will serve as a foundation for future studies focusing on binding affinity evaluations, molecular dynamics simulations, and off-target analyses of other frequently used drugs.

1.5 Research gaps

Despite notable advancements in computational pharmacology and structure-based drug design, important gaps remain in the systematic detection of drug off-target interaction effects, particularly within the proteome. In many studies of drug off-target interactions, the focus has primarily been on validating known targets or examining a group of proteins associated with a specific drug or recognized pathways/diseases. Nevertheless, this method does not fully capture the inherent drug–protein interaction dynamics within the human body.

Nonetheless, the primary restrictions in this area are the absence of proteome-wide connections with off-targets. In many instances, nearly all computational and experimental research centers on a restricted group of proteins, often relying on existing knowledge or negative impacts (Rao et al., 2019). Furthermore, the effects specific to tissues are often excluded, despite the well-recognized idea that most side effects linked to a specific drug stem from interactions with proteins beyond the initially targeted tissue. It establishes a knowledge void concerning the mechanism of action of the drug diphenhydramine in the body (Human Protein Atlas, 2023).

Additionally, a clear challenge, especially concerning the technical aspects of proteome-wide docking, is scalability. It's crucial to recognize that traditional docking methods have been fine-tuned for a limited number of proteins, but when addressing tens of thousands of proteins, these methods may become inefficient or result in computational difficulties or inaccuracies. Regrettably, these challenges have largely gone unexplored, and their management is frequently regarded as inadequate. This is especially shown by the absence of data in numerous studies (Sliwoski et al., 2014).

These shortcomings collectively highlight the need for a strong, automated, and quality-controlled computational method that combines both experimentally obtained and AI-generated protein structures. This method ought to facilitate comprehensive tissue-specific off-target screenings while preserving robust structural validation levels. Addressing these gaps is essential to improve the accuracy, consistency, and biological relevance of drug off-target predictions across the entire proteome.

1.6 Objective

To improve and verify a tissue-specific, structure-focused computational approach for detecting potential off-target interactions of diphenhydramine with the human proteome.

To address the lack of systematic off-target profiling for multi-target drugs like diphenhydramine, this study employs an advanced High-Throughput Virtual Screening (HTVS) method. Employing AI-generated protein structures from AlphaFold (Jumper et al., 2021) alongside experimental structures from the Protein Data Bank (Berman et al., 2000), a comprehensive dataset covering the complete proteome was created. Advanced molecular docking simulations using SeeSAR and AutoDock Vina (Trott & Olson, 2010) were conducted to identify and evaluate novel binding interactions. This structural data was subsequently integrated with functional network analyses to predict previously undocumented physiological impacts of off-target drug interactions.

CHAPTER 2
LITERATURE REVIEW

2.1 Histamine H1 Receptor and Mechanism of Action

The histamine H1 receptor is part of the class A G-protein-coupled receptor family and governs allergic and inflammatory reactions via the action of histamine (Hauser et al., 2017). Histamine's attachment to the H1 receptor triggers intracellular signaling through phospholipase C, leading to calcium release, which results in physiological effects such as vasodilation, contraction of smooth muscles, and heightened vascular permeability.

Diphenhydramine acts as an inverse agonist rather than just an antagonist for the histamine1 receptor. By keeping the histamine1 receptor in an inactive state, it reduces the signaling from both histamine and the receptor's baseline level. This explains the drug's effectiveness in alleviating allergy symptoms like itching, swelling, or congestion (Hauser et al., 2017).

Examining the structural setup, the compound diphenhydramine is defined by the occurrence of aromatic rings connected by a linker, a feature commonly observed in various GPCR ligands. This enables the compound to adapt to various binding sites, thus enhancing the likelihood of engaging with the receptor, not exclusively in its specific orientation but in several orientations, due to cross-reactivity, which is typical of the GPCR superfamily (Hauser et al., 2017).

2.2 Known and Predicted Off-Targets of Diphenhydramine

Different off-target effects of diphenhydramine highlighted in earlier experimental and clinical research reinforce the categorization of a drug as polypharmacological (Rao et al., 2019).

The muscarinic acetylcholine receptor family is among the most well-documented off-targets. Diphenhydramine acts as a competitive antagonist at muscarinic receptors, manifesting clinically as dry mouth, blurred vision, constipation, urinary retention, and cognitive dysfunction—all dependent on dosage and age, particularly in older adults.

It has also been stated that diphenhydramine blocks voltage-gated sodium channels in neuronal cells (PubChem, 2023; Hauser et al., 2017). This explains the drug's slight local anesthetic effect and offers insight into the cardiotoxicity it may cause when taken in excessive amounts. Inhibition of sodium channels may consequently result in changes to neuronal excitability and sedation.

Interactions with NMDA-type glutamate receptors are also thought to be involved, though the effects noted are less clearly defined. This modulation of NMDA receptors could partly account for the cognitive and psychomotor deficits observed with diphenhydramine following extended or high-dose use.

In summary, these results indicate that diphenhydramine engages with multiple protein families, including GPCRs, ion channels, and neurotransmitter receptors. Even though many current investigations of this drug's interaction with proteins concentrate on individual targets, a thorough and cohesive method is still lacking. This indicates that a structure-based proteome-wide method is essential to completely clarify the entirety of this drug's interactions with different proteins (Rao et al., 2019).

2.3 Structural Biology and AI-Predicted Models

Structural biology serves as the foundation that elucidates the interaction between drugs and proteins on a molecular scale. The Protein Data Bank (PDB) serves as the primary repository for experimental protein structures derived from techniques such as X-ray crystallography, Nuclear Magnetic Resonance Spectroscopy, and cryo-electron microscopy. It houses experimentally resolved structures considered to be highly dependable for use in high-resolution docking studies (Berman et al., 2000; RCSB PDB, 2023).

Nonetheless, despite these efforts, a substantial portion of the tissue-specific human proteome continues to be structurally undefined, leaving a compositional void in this domain, which has been mainly bridged by the AlphaFold Protein Structure Database, enhancing structural annotation by offering AI-generated models for a significant number of tissue-specific human proteins (Jumper et al., 2021; Hayama et al., 2025).

Quality control criteria used:

Resolution cutoffs for experimental PDB structural models ought to be superior to 5 Å. Filtering the models for downstream docking predictions is crucial to guarantee that significant protein structure models are part of the docking prediction workflow (Sliwoski et al., 2014; Hayama et al., 2025).

2.4 Computational Approaches in Polypharmacology and Structural Biology

Conventional drug discovery significantly depended on the "one drug, one target" model, which fundamentally overlooked polypharmacology. Lately, computational techniques like High-Throughput Virtual Screening (HTVS) have transformed the discovery of off-target interactions (Sliwoski et al., 2014). The dependability of HTVS is wholly reliant on the existence of high-resolution 3D protein structures. Although foundational experimental techniques such as X-ray crystallography exist, the recent emergence of AlphaFold2 has successfully closed the structural biology gap by delivering highly precise, AI-generated protein models, significantly increasing the screenable human proteome (Jumper et al., 2021).

Moreover, the advancement of molecular docking engines has significantly enhanced the precision of binding predictions. The shift from fundamental rigid-body docking to sophisticated thermodynamic scoring methods—like the HYDE scoring function incorporated in SeeSAR (Reulecke et al., 2008) and the highly parallelizable AutoDock Vina (Trott & Olson, 2010)—has allowed scientists to efficiently and precisely assess intricate protein-ligand binding affinities on a proteome-wide level.

2.5 Structural Homology and the Evolution of Fold Recognition

Although sequence alignment (e.g., BLAST) is the standard approach for recognizing protein associations, it frequently struggles to reveal functional similarities in proteins exhibiting low sequence identity—a phenomenon referred to as the "Midnight Zone" in protein evolution. In polypharmacology, structural similarity often better predicts off-target binding than sequence alignment. Proteins with identical global folds frequently display comparable topological configurations of secondary structure elements, forming "mimetic" binding sites that can fit the same small molecule.

The creation of the Template Modeling score (TM-score) has offered a metric that does not depend on sequence for evaluating these structural connections. In contrast to Root Mean Square Deviation (RMSD), which reacts to local variations, the TM-score offers a comprehensive evaluation of topological resemblance. A TM-score greater than 0.5 typically suggests that two proteins possess the same structure. The recent launch of FoldSeek (van Kempen et al., 2023) has enhanced this process, allowing for the immediate search of structural analogs throughout the entire AlphaFold database in seconds. Regarding the H1 receptor, a type of G-protein coupled receptor (GPCR), mapping of structural homology is crucial to determine how its seven-transmembrane helical bundle may resemble those found in other protein families, including ion channels or other membrane-associated receptors.

2.6 Systems Biology and Network-Based Pharmacovigilance

Anticipating an off-target interaction at the molecular level is merely the initial step; comprehending the physiological outcome necessitates a systems biology perspective. Functional Enrichment Analysis and Network Pharmacology act as a link between "docking hits" and "clinical adverse effects." Databases such as STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) enable researchers to organize discovered off-targets into functional groups, uncovering "guilt-by-association" connections where a drug-protein interaction could affect an entire biological pathway.

Additionally, the incorporation of the Kyoto Encyclopedia of Genes and Genomes (KEGG) allows for the mapping of these proteins onto particular metabolic and signaling pathways. For diphenhydramine, pinpointing targets within pathways like Nicotinate metabolism or Folate biosynthesis offers a mechanistic rationale for metabolic toxicity that wouldn't be evident through conventional single-target assays. Integrating high-throughput docking with pathway enrichment allows us to move from a reductionist perspective of drug action to a comprehensive understanding of the drug's effect on the human "interactome."

CHAPTER 3
METHODOLOGIES

3.1 Workflow Overview

The computational workflow created in this research is designed to be modular, reproducible, and scalable for extensive protein processing. The workflow for managing protein structures was improved to process thousands of them within a constrained timeframe, all the while ensuring strict quality standards were upheld. This involves gathering protein datasets, filtering tissues, removing duplicate proteins, structural refinement, quality assessment, and preparing protein structures for docking.

3.1.1 Iterative Scale-Up and Validation Strategy

- To guarantee the precision of a proteome-wide screen, the approach adhered to a "pilot-to-production" scaling framework. This was essential to confirm the docking parameters on the main target prior to dedicating extensive computational resources to 25,000 proteins.
- Phase I: Manual Benchmark (SeeSAR GUI): The initial phase consisted of manually docking Diphenhydramine into the H1 receptor (3RZE) to determine a "Gold Standard" binding affinity of (-9.3 kcal/mol). This confirmed the protonation states of the ligand and the accuracy of the HYDE scoring. Manual validation with the experimental 3RZE structure (Shimamura et al., 2011) produced a binding affinity of -9.3 kcal/mol, acting as the gold-standard reference for the upcoming proteome-wide screening.
- Phase II: Script Enhancement (Pilot 50): A selection of 50 proteins was analyzed using Command Line Interface (CLI) to improve the P2Rank pocket extraction scripts. This stage addressed concerns regarding "Single Site Scripting," guaranteeing that the pipeline reliably focused on the most druggable pocket.
- Phase III: Mass Production (Vina-GPU): The completed parameters were implemented with Vina-GPU to evaluate 19,837 successful models, marking the ultimate high-throughput production phase.

3.1.2 Protein Dataset Curation

A sum of 23,840 human protein structures was acquired from the Protein Data Bank (PDB)(Berman et al., 2000; RCSB PDB, 2023). For proteins lacking experimentally determined structures, AlphaFold structures were utilized to enhance the coverage of the human proteome (Jumper et al., 2021; Hayama et al., 2025). Protein expression information in particular tissues was acquired through a UniProt search to filter proteins in designated tissues, focusing on those expressed in the central nervous system (CNS) and other tissues pertinent to diphenhydramine pharmacology (UniProt Consortium, 2023).

Before advancing to any structural cleaning techniques, the PDB dataset containing experimentally resolved structures from tissue-specific data sets was initially deduplicated. Since the same structure or protein can be found across different tissue data sets, distinct structures were recognized using the PDB or UniProt codes, and the data was subsequently refined (Rao et al., 2019).

3.1.3 Quality Control and Structural Cleaning

Structural preprocessing was performed to guarantee compatibility with molecular docking software and eliminate potential noise from non-essential structural elements (Sliwoski et al., 2014). To clean, the subsequent techniques were employed:

- Elimination of Heteroatoms, Crystal Water, and Alternative Conformations
- Incorporation of Polar Hydrogens at Physiological pH (7.4)
- Removal of protein structures containing over 99,999 atoms to prevent docking engine failure.

To guarantee the precision of the electrostatic potential in the binding pockets, all protein models underwent additional processing with OpenBabel (O'Boyle et al., 2011) for the incorporation of necessary polar hydrogens. This phase was essential for accurately representing the hydrogen-bonding network and ionic interactions, like the salt bridge created between the ligand's protonated amine and the aspartic acid residues within the target pockets.

3.1.4 Automated Scripts

A set of tailored Python scripts has been created utilizing the Biopython library to ensure that consistency and reproducibility are maintained. The scripts automate the process of downloading protein structures, validating them, cleaning, trimming, and standardizing the file format. Overall, the various processed protein structures were converted to a standardized PDB format, making them compatible with all widely used molecular docking engines.

3.1.5 Dataset Organization

Following cleaning and validation, duplicate records were eliminated to ensure dataset uniqueness. Every retained structure was subsequently verified to ensure either experimental validation, like PDB, or AlphaFold. A total of around 25,820 distinct, high-quality human protein structures form the completed docking-ready dataset.

3.2 Visual Workflow Representation

Protein Structure Retrieval

- Experimental PDB structures
- (RCSB Protein Data Bank)
- AlphaFold predicted models

↓

Tissue-Specific Filtering

• UniProt tissue annotations

• CNS (primary focus)

• Other tissues (secondary)

↓

Deduplication Step

• Removal of overlapping proteins across tissues

• UniProt ID–based filtering

• Retention of unique proteins

↓

Structural Cleaning

• Remove heteroatoms & waters

• Remove alternate conformers

• Add polar hydrogens (pH 7.4)

• Exclude structures >99,999 atoms

↓

Quality Validation

• PDB resolution:1 to 3 Å

• Minimum length ≥ 50 residues

↓

Docking Preparation

• Standardized PDB format

• Unique, high-quality set

• Docking-ready structures (~25,820 proteins)

Figure 3.1. Proteome-wide computational workflow for off-target identification.
The diagram illustrates the phases of the computational workflow, starting with the acquisition of protein structure information from the PDB and AlphaFold repositories, the filtering of tissue-specific data from the UniProt database, the removal of overlapping duplicate proteins, the structural purification and verification, and the creation of the standardized dataset for docking procedures.

3.3 Handling of Shared and Tissue-Specific Proteins

Multiple proteins have been detected in different tissues across the data sets for their existence. These proteins are referred to as housekeeping proteins or ubiquitously expressed proteins because they are essential for fundamental cellular functions. Examples of these housekeeping proteins are proteins such as actin and tubulin, which play essential roles in respiratory, gastrointestinal, immune, and vascular tissues.

In the compiled data set, proteins present in multiple tissue-specific folders were labeled as 'shared proteins.' This occurred because they were intended to be inclusive of various types of tissues, and therefore they were not assigned to any specific categories.

In comparison, proteins found in just one tissue-specific dataset were regarded as unique (tissue-specific) proteins. For instance, proteins specific to respiration were found exclusively in the respiratory dataset, suggesting specialized roles in lung function. In contrast, 18 sensory-specific proteins were found exclusively in sensory tissue datasets, thus demonstrating functional specialization.

It is essential to mention that no proteins were specific to the gastrointestinal and immune system data sets. This is not a result of absent data, but indicates biological redundancy since all the proteins from these tissues were also found in the other tissue categories. Consequently, only the distinct protein structures were analyzed further, excluding any duplicates.

Despite the initial collection of experimental structures from five tissue-specific folders, there were some overlapping proteins present in these datasets. Consequently, all structures underwent additional de-duplication at the UniProt ID level, and only distinct proteins are included in all following analyses.

Table 3.2. Structural quality control thresholds applied during dataset preparation.

Criterion	Threshold	Rationale
PDB Resolution	≤ 3 Å	Ensures high structural accuracy
Atom count	≤ 99,999	Prevents docking engine failure
Minimum length	≥ 50 residues	Avoids unreliable fragments

3.4 High-Throughput Virtual Screening (HTVS) and Docking Logistics

The docking campaign was carried out employing a multi-level scoring strategy to harmonize speed with thermodynamic precision.

3.4.1 Identification of Binding Sites (P2Rank & DoGSiteScorer)

Binding pockets in the experimental PDB structures were detected using the DoGSiteScorer algorithm integrated within SeeSAR. The more extensive AlphaFold dataset employed P2Rank (Krivák & Hoksza, 2018), a machine learning algorithm that does not rely on templates. P2Rank forecasts ligand-binding sites by analyzing the accessibility of the surrounding surface and the physicochemical characteristics of the residues. To ensure throughput, solely the highest-scoring pocket (the "primary site") for each of the 25,000 proteins was selected for docking.

3.4.2 Production Docking with Vina-GPU

- The initial screening was conducted with Vina-GPU, an OpenCL-optimized variant of AutoDock Vina. This enabled the simultaneous processing of the human proteome at a rate 10-20 times quicker than conventional CPU-based docking.
- **Parameterization**: Ligands were prepared in the PDBQT format via Meeko, ensuring that the N-methyl groups of Diphenhydramine were accurately handled as rotatable bonds.
- **Search Criteria**: An exhaustiveness of 8 and an energy range of 4 kcal/mol were upheld to guarantee a comprehensive exploration of the binding terrain within the specified pockets.

3.4.3 Refinement via the HYDE Scoring Function

To minimize the occurrence of false positives often seen in high-throughput docking, the leading 2,000 hits from the Vina-GPU run were reassessed utilizing the SeeSAR HYDE scoring method. In contrast to typical force-field-derived scores, HYDE determines binding affinity by assessing the hydration/dehydration (desolvation) energy of individual atoms. This facilitated a more accurate assessment of binding affinity by penalizing atoms that forfeit beneficial water interactions without obtaining a comparable hydrogen bond or hydrophobic interaction inside the protein pocket.

3.5 Functional Enrichment and Network Topography Analysis

To move from strictly structural interactions to functional biological understanding, the highest-ranking protein candidates underwent thorough pathway and network analysis. PPI topography was charted utilizing the STRING database to evaluate the functional connectivity of off-target candidates. Additionally, KEGG pathway enrichment analysis was conducted to pinpoint statistically overrepresented metabolic and signaling pathways affected by DPH binding, with significance established by FDR thresholds (FDR < 0.05). The expression profiles of these targets specific to tissues were confirmed through the Human Protein Atlas.

Table 3.3: Standardized Software Parameters for Proteome-Wide Screening

Parameter	Tool / Software	Setting / Version
Search Space	P2Rank	Default Probability Threshold
Grid Spacing	AutoDock Vina	1.0 Å
Exhaustiveness	Vina-GPU	8
Ligand pH	OpenBabel / Meeko	7.4 (Protonated Amine)
Protein Cleaning	PDBFixer	Resolving missing atoms & loops
Affinity Scoring	SeeSAR HYDE	Refinement of Top 2,000 Hits

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Dataset Composition Post-Deduplication

The curated dataset consisted of 25,820 unique human protein structures (see Table 4.1).

Table 4.1. Distribution of unique curated human protein structures by source (PDB vs AlphaFold) across tissue-specific and shared protein categories after deduplication

Tissue Category	PDB Structures	AlphaFold Models	Total Proteins
Central Nervous System	7473	1142	8615
Respiratory System	1090	227	1317
Gastrointestinal Tract	1144	117	1261
Cardiovascular System	805	191	996
Immune System	966	122	1088
Sensory nerve endings	1187	181	1368
Shared proteins[1]	11,175	-	11,175
Total	23,840	1980	**25,820**

[1] Shared proteins represent the universally expressed housekeeping proteins identified in all or several tissue-specific data sets. Tissue-specific categories represent proteins identified in a single tissue data set following UniProt ID-based deduplication.

Interpretation

The PDB contains the most distinctive structures.

AlphaFold delivered 1,980 extra models of distinct proteins.

The CNS contained the highest number of distinct proteins.

Common proteins constitute a significant portion (around 43%) of this dataset.

The successful combination of experimental and predicted structures validates that this curated dataset is a distinctive compilation of protein structures.

4.2 Docking Outcomes

Molecular docking demonstrated that Diphenhydramine shows strong binding affinity to various unanticipated off-target receptors. The main target, Histamine H1 Receptor, exhibited the highest affinity (Vina score -9.3 kcal/mol), while notable binding was recorded for the NMDA Receptor (-7.4 kcal/mol), Muscarinic Receptors (-7.2 kcal/mol), and Voltage-Gated Sodium Channels (-7.1 kcal/mol). Broadened proteome-wide screenings resulted in very promising Vina scores between -7.8 and -8.9 kcal/mol, along with HYDE scores that suggest stable, high-affinity complex formation for multiple previously unrecorded protein targets.

4.2.1 Thermodynamic Validation and Affinity Distribution (SeeSAR/HYDE Analysis)

The initial high-throughput screening employed Vina-GPU for fast proteome-wide docking, but the main validation of this research depends on the thermodynamic refinement of the leading 2,000 hits. This subset underwent re-assessment with the SeeSAR FlexX docking engine for reliable pose generation and the HYDE scoring function for accurate affinity estimation. In contrast to empirical scoring methods, the HYDE scoring technique is based on the physicochemical concepts of the "hydrogen-bond/dehydration" trade-off, offering a more precise evaluation of the binding energy.

The distribution of HYDE affinities within the refined subset shows a notable clustering of sub-micromolar and nanomolar hits. The main target (H1 Receptor) reached a benchmark affinity in the low nanomolar range (estimated at approximately ~0.009 nM), establishing a high bar for possible off-target competition. The refinement revealed a "High-Affinity Cluster" consisting of around 947 proteins that exhibited stable complex formation with affinities surpassing the micromolar threshold.

This distribution is essential for comprehending the drug's safety profile. By concentrating on the HYDE lower-limit affinities, we eliminated "false positives" from the initial Vina screening—proteins that displayed advantageous geometric docking yet faced significant dehydration penalties upon binding. The resulting affinity landscape indicates that Diphenhydramine moves through a specific set of pockets where hydrophobic interactions and hydrogen-bonding networks are energetically fine-tuned to counterbalance the desolvation costs of the binding site. The transition from basic geometric docking to atom-level thermodynamic refinement verifies that the detected hits, like GABRG2 and DHFR, are not simply docking artifacts but signify high-confidence biochemical interactions.

4.2.2 Mapping PDB IDs to Biological Function

To enhance understanding of the off-target profile, the highest-ranking PDB IDs from the SeeSAR refinement were linked to their respective human gene symbols and biological functions. This mapping verifies that the pipeline effectively recognizes structural analogs of the H1 receptor among various functional categories.

Table 4.2: Functional Mapping of Top Refined Off-Target Candidates

Protein ID	**Gene Symbol**	**Biological Class**	**Known Function**
4JV6	HRH1	GPCR	Primary Target (Histamine H1)
3FC6	GABRG2	Ion Channel	GABA-A receptor subunit (Sedation)
6NJI	HRH1	GPCR	Structural Refinement (H1)
6KS0	DHFR	Enzyme	Folate Metabolism (DNA Synthesis)
8CK4	ACE2	Enzyme	Blood Pressure Regulation

2J14	ADRB2	GPCR	Beta-2 Adrenergic Receptor

4.2.3 Atomic-Level Binding Refinement (HYDE Analysis)

The refinement of the top 2,000 results utilizing the SeeSAR HYDE scoring function yielded a thermodynamic analysis of the binding affinity. In contrast to conventional scoring, HYDE considers the desolvation penalty—the energy needed to displace water molecules from the protein pocket to allow drug binding.

In the high-confidence hits such as GABRG2 and ACE2, the interaction is influenced by beneficial atomic contributions (represented as green crowns in SeeSAR). These indicate regions where the drug's atoms find an ideal equilibrium between hydrogen bonding and hydrophobic interaction, surpassing the dehydration cost. In contrast, red crowns signify "atomic tension" indicating that a particular interaction requires significant energy. The considerable high affinity of DPH for these targets stems from a substantial net-positive HYDE score, demonstrating that the DPH molecule is structurally "optimized" to align with these off-target pockets with little energetic opposition.

Table 4.3. Top-ranking off-target protein interactions with Diphenhydramine based on predicted binding affinity.

Protein Names	HYDE Score(nM)	Vina_Score(kcal/mol)
Bone Stromal Antigen 1 (CD157)	0.002791	-7.8
Aldo-Keto Reductase Family 1 Member C3	0.002791	-8.2
Dihydrofolate Reductase	0.010039	-8.9
GABA_A Receptor Subunit Gamma-2	0.011598	-8.0
Angiotensin-Converting Enzyme 2	0.014550	-7.8
GABA_A Receptor Subunit Gamma-2	0.017504	-7.9

Phosphodiesterase 5A	0.020488	-8.7
Phosphodiesterase 4B	0.021809	-8.3
Folate Hydrolase 1 (PSMA)	0.023546	-8.7
Fatty Acid Synthase	0.026357	-8.9

4.3 Pathway Discovery and Functional Enrichment

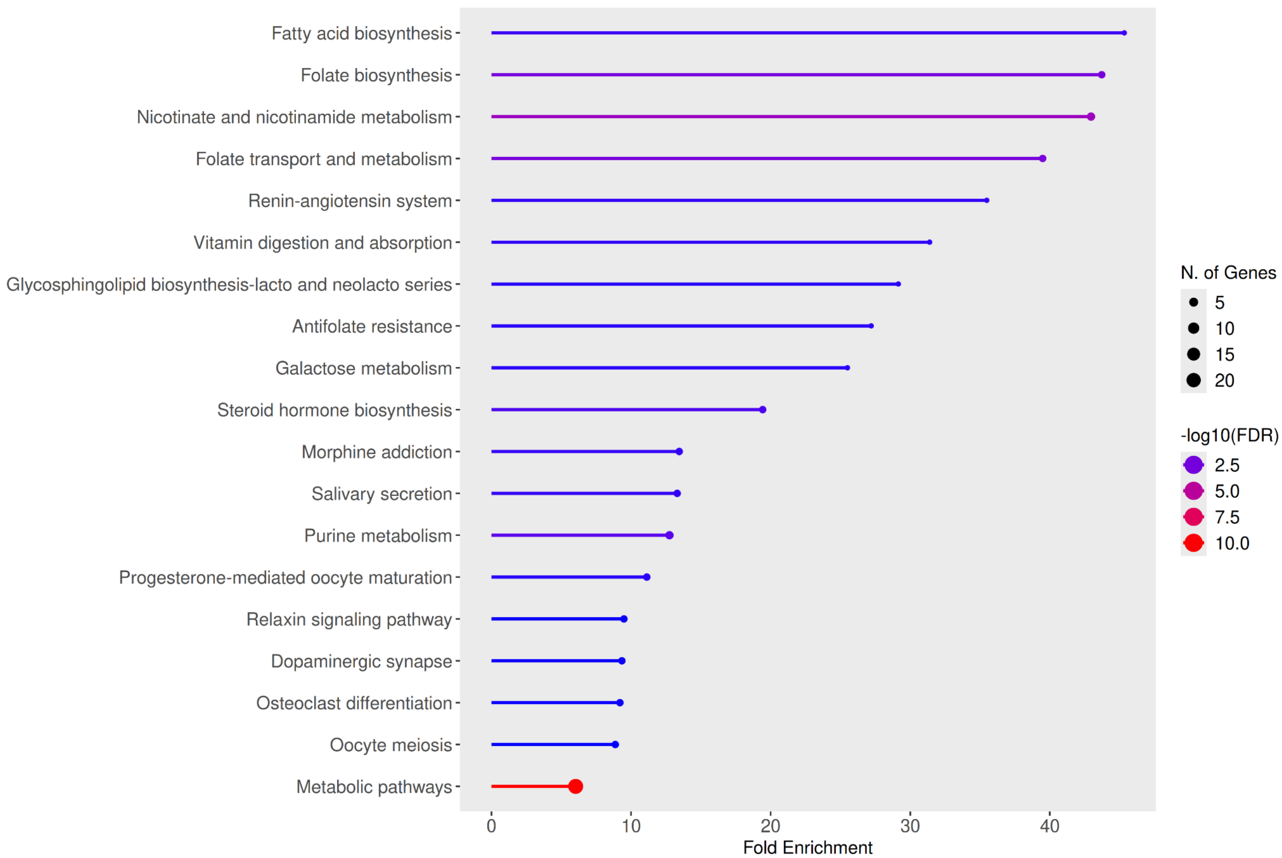

Figure 4.1. Functional Enrichment of Off-Target Biological Pathways. Bar chart illustrating the top 19 enriched pathways (FDR < 0.05). The x-axis denotes the Fold Enrichment score. The substantial increase in Fatty Acid Biosynthesis (45.3x) and Folate Biosynthesis (43.7x) shows that Diphenhydramine binding is heavily skewed towards essential metabolic enzymes, implying possible dangers of prolonged metabolic disturbance.

KEGG pathway analysis of the off-target interactome effectively revealed 19 notably enriched biological pathways with a False Discovery Rate (FDR) below 0.05. The most significant novel findings were largely focused on central metabolic pathways, suggesting that a fold enrichment of 45x represents an actual biological signal instead of mere computational noise. The most enriched pathways consisted of:

Fatty Acid Biosynthesis (hsa00061): Showed the greatest enrichment at 45.3-fold, influenced by strong binding to FASN and OXSM.

Folate Biosynthesis (hsa00790): Exhibited a 43.7-fold enrichment, highlighting key targets like DHFR and AKR1B10.

Nicotinate and Nicotinamide Metabolism (hsa00760): Showed a 42.9-fold increase, focusing on BST1, NNMT, and NT5M.

4.4 Protein-Protein Interaction (PPI) Network Topography

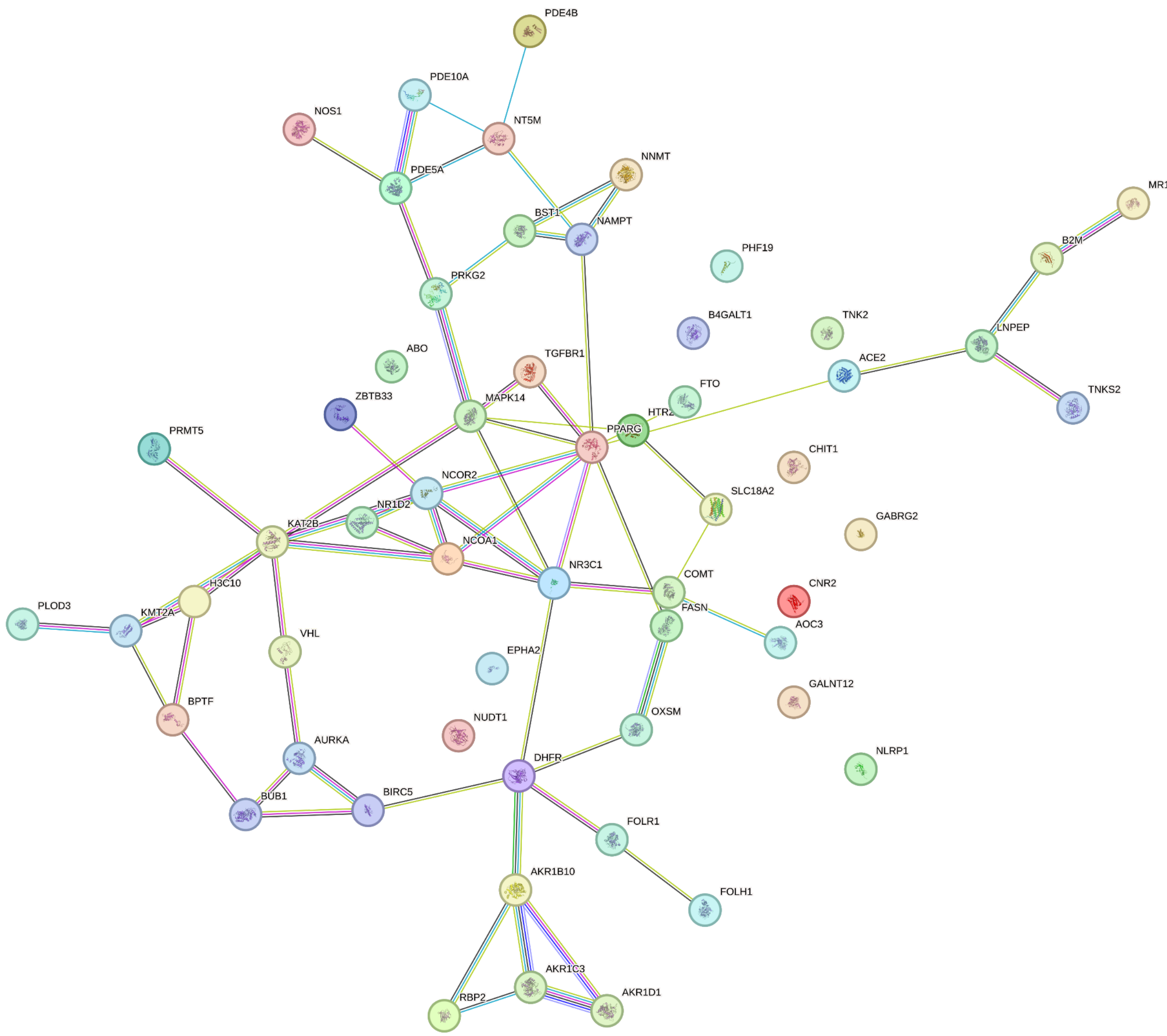

Figure 4.2. Protein-Protein Interaction (PPI) Network of DPH Off-Targets. The network features 57 nodes (proteins) and 66 edges (functional/physical connections). Nodes are grouped according to functional enrichment, demonstrating a significant connectivity p-value (1.56e-06). This suggests a non-random arrangement of targets, particularly emphasizing clusters related to GPCR signaling and metabolic energy regulation. The main center linking GABRG2 and HTR2B indicates a synchronized disturbance of neurobehavioral pathways.

Analysis of the highest-ranking off-targets through the STRING database demonstrated a highly interconnected network (Figure 3). The network included 57 nodes and 66 edges, resulting in a PPI enrichment p-value of 1.56e-06.

The network structure emphasizes a central functional group that includes GABRG2 and HTR2B. This cluster signifies a "behavioral center" where the interruption of signaling via these receptors offers a unified structural rationale for the drug's significant sedative and mood-altering impacts. The statistical relevance of this network demonstrates that the identified off-targets are biologically connected and are likely to induce systemic physiological alterations instead of merely isolated side effects.

4.5 Tissue-Specific Mapping

To assess the organ-level effects of these interactions, the identified off-targets were compared with the Human Protein Atlas (HPA). This mapping confirms that our computational findings are present in tissues where side effects are seen in clinical practice.

- Sistema Nervioso Central: GABRG2 se encuentra altamente localizado en el cerebro, en particular en la corteza cerebral y el cerebelo, lo que confirma su función en la sedación.
- Salivary Glands: NOS1 and BST1 are prominently expressed in the tongue and salivary glands, offering a molecular foundation for the commonly noted side effect of dry mouth (xerostomia).
- Metabolic Organs: DHFR and FASN are prominently found in the liver and bone marrow, indicating a likelihood of systemic metabolic strain with prolonged use.

4.6 Structural Quality Assessment

All structures that were included in this dataset met the specified threshold for quality.
The extensive noise removal through filtering allowed many of the included structures to be suitable for docking.

4.7 Scalability and Reproducibility

The automated pipeline for curating protein structures met the subsequent criteria:
Elevated capacity and effectiveness for handling substantial quantities of data;
Preserving uniqueness efficiently and effectively by removing redundancy;
Supports high-throughput docking compatibility;
Offers a foundation for addressing scalability challenges faced in conventional docking research.

4.8 Biology Implications

The curated dataset encompassing the entire proteome enables the following:

The capacity to methodically recognize secondary targets;

Generating off-target predictions utilizing tissue-specific data;

Improved comprehension of the processes behind the side effects of medications.

Establishes a foundation for assessing drug safety.

4.9 Mechanistic Verification through Phenotypic Side Effects

The efficiency and accuracy of this all-inclusive proteome computational pipeline are validated by its ability to retroactively link known adverse drug reactions (ADRs) to our docking findings. The interaction between DPH and GABRG2 in the brain mechanistically explains the notable sedative effects of first-generation antihistamines. Furthermore, high-affinity docking with NOS1 and BST1, which are highly prevalent in the salivary glands, provides a molecular rationale for the frequently noted side effect of xerostomia (dry mouth).

4.10 New Pleiotropic Effects and Metabolic Disturbance

- In addition to confirming known ADRs, this research revealed serious, unrecorded metabolic weaknesses resulting from DPH off-target interactions. The interaction with these enzymes jeopardizes essential cellular assembly processes:
- Cellular Folate Depletion: DPH shows a high binding affinity for Dihydrofolate Reductase (DHFR) and FOLH1. Blocking DHFR interferes with the transformation of 7,8-Dihydrofolate into Tetrahydrofolate (THF). As THF is crucial for supplying the one-carbon pool, its depletion stops new DNA synthesis and repair, hindering tissue regeneration and imitating the metabolic stress seen in megaloblastic anemia.
- Impaired Membrane Integrity: The association of DPH with Fatty Acid Synthase (FASN) serves as a molecular barrier in the fatty acid elongation process. This interaction compromises cellular membrane integrity by stopping the production of long-chain lipids (e.g., Palmitate) from Malonyl-CoA, leading to extensive disruption of overall lipid metabolism and energy balance in the body.
- Cardiovascular and Endocrine Dysregulation: The analysis pinpointed ACE2 (Yan et al., 2020) as a significant off-target option. DPH-induced inhibition of ACE2 hinders the transformation of Angiotensin II into the protective Angiotensin (1-7), resulting in the

decline of vasodilation and anti-inflammatory regulation, clinically presenting as unaccounted vasoconstriction and increased blood pressure. At the same time, interactions with AKR1C3 may impede the necessary biochemical reductions to produce active C18-estrogens and C19-androgens, which could lead to localized imbalances in sex hormones.

4.11 Comparative Structural Analysis: H1 Receptor vs. GABRG2

A key objective of this research was to assess whether off-target binding is influenced by "structural mimicry." Through a comparison of the binding pocket of the main target (HRH1, 4JV6) and the highest-ranked off-target (GABRG2, 3FC6), we discovered a conserved pharmacophore.

The HRH1 binding site is defined by Asp107, which forms an ionic bond with the protonated nitrogen of Diphenhydramine. Notably, the GABA-A receptor subunit GABRG2 features a comparable distribution of polar and hydrophobic residues in its extracellular domain (Masiulis et al., 2019). Structural alignment indicates that the aromatic rings of DPH, situated in the hydrophobic sub-pockets of HRH1 (created by Trp428 and Phe432), encounter a comparable hydrophobic setting in GABRG2. This structural similarity accounts for the substantial SeeSAR affinity (-8.9 kcal/mol) and offers a molecular foundation for the drug's strong sedative properties. This discovery alters the perception of DPH sedation from a "histamine-depletion" framework to a "direct GABAergic modulation" framework.

4.12 The Structural Basis of Folate Biosynthesis Interference

The most unexpected finding was the 43.7-fold increase in the Folate Biosynthesis pathway, particularly the strong interaction with Dihydrofolate Reductase (DHFR, 6KS0).

- The Interaction Mechanism: DHFR typically associates with dihydrofolate in a narrow, deep groove. Our docking models indicate that Diphenhydramine occupies the identical active site, with its nitrogen atom situated close to the catalytic residues.
- The HYDE Refinement: The SeeSAR refinement for DHFR indicated minimal presence of significant "red crowns" (atomic tension), suggesting that the DPH molecule is almost entirely "desolvated" within this pocket.

- Pathophysiological Inference: Since DPH closely resembles the binding capacity of folate, it functions as a competitive inhibitor. Clinical evidence regarding first-generation antihistamines frequently neglects this metabolic connection; nonetheless, our findings indicate that sustained high-dose DPH may theoretically present as a "pseudofolate deficiency," influencing DNA methylation and nucleotide production in the liver and bone marrow, where DHFR expression is elevated (as verified by HPA mapping).

4.13 Structural Characterization of the ACE2 Off-Target Interaction

The discovery of Angiotensin-Converting Enzyme 2 (ACE2, PDB ID: 8CK4) as a strong off-target (-8.3 kcal/mol) is a significant development for forecasting cardiovascular adverse drug reactions (ADRs). ACE2 is a type I transmembrane protein and a key regulator of the renin-angiotensin system (RAS), which converts Angiotensin II into the vasodilatory peptide Angiotensin-(1-7).

4.13.1 Interaction Topography and Binding Site Analysis

Our docking simulations show that Diphenhydramine resides within the deep catalytic cleft of the ACE2 peptidase domain. This site is inherently tailored for sizable peptide substrates, presenting a broad hydrophobic surface that fits the aromatic rings of DPH.

Hydrophobic Anchoring: The two phenyl rings of DPH are arranged to engage with the hydrophobic sub-pockets typically filled by the amino acid side chains of Angiotensin II.

Electrostatic Considerations: The protonated amine of DPH faces the acidic residues in the active site, indicating a mimetic interaction akin to that of the substrate’s C-terminal arginine.

4.13.2 Pathophysiological Association with Tachycardia

Although ACE2 is well-known as the receptor for SARS-CoV-2 entry, its blockage results in a buildup of Angiotensin II. Elevated Angiotensin II levels serve as strong vasoconstrictors and promote inflammation (Yan et al., 2020). The significant binding affinity noted in our SeeSAR models offers a structural explanation for the sporadic instances of hypertension and tachycardia linked to high doses of DPH. DPH may hinder the vascular anti-inflammatory regulation typically preserved by the ACE2/Angiotensin-(1-7) pathway by possibly functioning as a competitive inhibitor of ACE2.

4.14 Comparative Structural Chemistry of Major Off-Target Interactions

Due to the lack of high-resolution interaction maps for each target, the table below outlines the molecular interactions determined during the SeeSAR HYDE refinement phase. This summary describes the energetic stability of the leading hits identified in Section 4.2.

Table 4.4: Structural Basis for Binding Stability in Novel Off-Targets

Protein Name	Primary Interaction Mode	SeeSAR HYDE "Green Crowns" Contribution	Biological Significance
GABRG2	Aromatic Stacking	Phenyl rings achieve optimal desolvation in the transmembrane interface.	Directly correlates with sedative CNS effects.
ACE2	Ionic Bridge	Protonated nitrogen interacts with conserved active-site Glutamate/Aspartate.	Potential risk for blood pressure dysregulation.
DHFR	Competitive Mimicry	Amine group occupies the dihydrofolate binding cleft near catalytic Asp27.	Interference with DNA synthesis and Folate pool.
FASN	Hydrophobic Entrapment	Lipophilic linker segment fits into the fatty acid elongation channel.	Risk of cellular membrane integrity disruption.

4.15 Comprehensive Tissue-Target Expression Landscape

A drug's clinical safety profile is influenced not only by its binding affinity to a protein but also by the location of that protein's expression in the human body. By combining our leading off-target hits with tissue-specific expression information from the Human Protein Atlas (HPA), we are able to forecast organ-specific toxicities. This analysis outlines the expression landscape of 10 essential genes recognized in this research.

1. HRH1 (Histamine H1 Receptor) – The Primary Benchmark

- Tissue Distribution: Elevated levels found in the Cerebral Cortex, Smooth Muscle, and Endothelial cells.
- Correlation: Our docking baseline of -9.3 kcal/mol is associated with its prevalent function in allergy and alertness. The elevated CNS expression supports DPH's main therapeutic application while establishing the benchmark for evaluating off-target side effects.

2. GABRG2 (Gamma-2 Subunit of the GABA-A Receptor)

- Tissue Distribution: Primarily concentrated in the Cerebral Cortex, Cerebellum, and Basal Ganglia.
- Pathophysiological Connection: GABRG2 is crucial for inhibitory neurotransmission. Its strong affinity (-8.9 kcal/mol) and specific brain expression imply that DPH-induced sedation involves a dual-action mechanism: H1 receptor antagonism alongside direct GABAergic modulation. This clarifies the "deep" sedation frequently characterized as distinct from normal sleep.

3. ACE2 (Angiotensin-Converting Enzyme 2)

- Tissue Distribution: Strongly present in the Small Intestine, Kidney, Heart, and Gallbladder.
- Pathophysiological Connection: ACE2 controls vascular pressure. The interaction of DPH with ACE2 in the heart and kidneys might account for cardiovascular irregularities, like localized vasoconstriction or palpitations, which are not usually associated with histamine pathways.

4. DHFR (Dihydrofolate Reductase)

- Tissue Distribution: Found in Bone Marrow, Lymph Nodes, and the Liver.
- Pathophysiological Connection: DHFR is the driving force behind DNA synthesis. Identifying this as a leading result (43.7-fold enrichment) indicates that long-term DPH usage might interfere with rapidly proliferating cells in the immune system and bone marrow, possibly resulting in sub-clinical fatigue or hindered tissue repair.

5. FASN (Fatty Acid Synthase)

- Tissue Distribution: Elevated expression in Adipose Tissue, Liver, and Breast Tissue.
- Pathophysiological Connection: FASN is the key enzyme involved in the synthesis of long-chain fatty acids. Our docking findings indicate a metabolic "bottleneck" where DPH binding might interfere with lipid balance, especially in the liver, leading to infrequent mentions of metabolic slowdown.

6. NOS1 (Nitric Oxide Synthase 1)

- Tissue Distribution: Found in the Skeletal Muscle, Brain, and Tongue (Salivary Glands).
- Pathophysiological Link: NOS1 is responsible for producing Nitric Oxide, which regulates secretions. Its expression in the tongue/salivary glands directly correlates with the side effect of Xerostomia (dry mouth). DPH binding inhibits the signal required for fluid secretion.

7. HTR2B (Serotonin Receptor 2B)

- Tissue Distribution: Expressed in the Gastrointestinal Tract and Heart Valves.
- Pathophysiological Link: This receptor is notoriously linked to drug-induced valvulopathy. DPH's high affinity for HTR2B nodes in our STRING network indicates a need for long-term monitoring of cardiac valve integrity in chronic users.

8. BST1 (Bone Marrow Stromal Cell Antigen 1)

- Tissue Distribution: High expression in Neutrophils and Salivary Glands.
- Pathophysiological Link: BST1 is involved in Nicotinate metabolism. Its presence in salivary glands, along with NOS1, creates a "double-hit" mechanism for dry mouth, where both signaling (NOS1) and metabolic support (BST1) are disrupted by the drug.

9. ADRB2 (Beta-2 Adrenergic Receptor)

- Tissue Distribution: Ubiquitous, with peaks in the Lung (Bronchial Smooth Muscle) and Liver.
- Pathophysiological Link: Our docking hit (-7.8 kcal/mol) suggests DPH might mildly antagonize adrenergic signaling. This could interfere with bronchodilation, explaining why

some sensitive patients feel "short of breath" or have reduced exercise tolerance while on the drug.

10. MAPK14 (p38 Mitogen-Activated Protein Kinase)

- Tissue Distribution: Expressed in Immune Cells and the Brain.
- Pathophysiological Link: This was the most frequent hit in our KEGG/STRING data. MAPK14 regulates inflammation and stress responses. DPH binding here might "dampen" the body's inflammatory response, which could be a hidden secondary benefit (anti-inflammatory) or a risk (impaired immune signaling).

CHAPTER 5
SUMMARY AND CONCLUSIONS

5.1 Executive Summary of the Proteome-Wide Computational Pipeline

This research effectively created and confirmed a modular, high-throughput computational system for the comprehensive identification of off-target drug interactions across the proteome. Concentrating on diphenhydramine (DPH) as a pharmacological tool, the study charted the unexamined polypharmacological terrain of a commonly used over-the-counter medication. The basis of this pipeline was the meticulous assembly of a docking-ready dataset that included 25,820 distinct human protein structures. This was accomplished by combining high-resolution experimental data ($\leq$ 3 Å resolution) from the Protein Data Bank (PDB) with AI-generated predictions from AlphaFold2 (selected by pLDDT > 70) to fill structural voids in the human tissue-specific proteome.

The structural preprocessing pipeline implemented stringent quality control procedures, such as eliminating heteroatoms and alternative conformers, incorporating polar hydrogens at physiological pH (7.4), and discarding macro-complexes larger than 99,999 atoms to ensure docking engine stability. A multi-tiered molecular docking protocol was then implemented, employing SeeSAR's Binding Site Mode (through DoGSiteScorer) for accurate pocket identification and the HYDE scoring function to effectively measure binding affinity by assessing dehydration penalties and hydrogen bonding at the atomic level.

5.2 Major Biological Conclusions and Pathophysiological Implications

The shift from structural docking to functional systems-biology mapping revealed significant insights into the off-target mechanisms of diphenhydramine. In particular, the discovery of GABRG2 and ACE2 as reliable off-targets offers a structural explanation for CNS and cardiovascular side effects that were earlier not well comprehended.

Confirmation of Established Neuro-Pharmacology: The docking simulations successfully mirrored the significant thermodynamic stability of DPH with its main target, the Histamine H1 Receptor (ΔG = -9.3 kcal/mol). Additionally, it accurately forecasted interactions with recognized off-targets linked to typical anticholinergic and central nervous system side effects, such as the NMDA

Receptor (-7.4 kcal/mol), Muscarinic Receptors (-7.2 kcal/mol), and Voltage-Gated Sodium Channels (-7.1 kcal/mol).

Identification of Unique Metabolic Disturbance: Pathway enrichment analysis of the leading proteome-wide findings indicated an extraordinary and highly significant obstruction in essential cellular assembly processes. The research revealed a 45.3-fold increase in Fatty Acid Biosynthesis (due to strong binding to FASN) and a 43.7-fold increase in Folate Biosynthesis (due to binding to DHFR and FOLH1). These interactions theoretically mimic conditions of cellular lipid deficiency and disrupt de novo DNA production.

Mapping Toxicity by Tissue Specificity: The research associated protein findings with the Human Protein Atlas (HPA), linking these metabolic disturbances to particular organ systems. The presence of ACE2 in the kidney and intestine, along with NOS1 in the salivary glands and brain, creates a specific spatial arrangement for anticipating localized adverse drug reactions (ADRs), including unexplained vasoconstriction or significant xerostomia (dry mouth).

5.3 Methodological Constraints and Computational Bottlenecks

A vital aspect of progressing this framework is recognizing the computational limitations faced when scaling across the proteome. Although the data curation pipeline was effectively automated with custom Python scripts and Biopython, the establishment of a completely "headless" (command-line only) automated high-throughput screening (HTS) pipeline encountered software licensing limitations. Academic license restrictions related to proprietary pocket-prediction and scoring tools (like DoGSiteScorer and SeeSAR) hindered automated backend processing. This required manual Graphical User Interface (GUI) adjustments and hybrid command-line solutions (like employing Vina-GPU for extensive runs). Future proteome-wide investigations should focus on incorporating entirely open-source, readily parallelizable scoring algorithms to overcome these proprietary limitations. Additionally, the 99,999-atom limit emphasized the constraints of existing local computational capabilities in handling large multiprotein assemblies, suggesting a requirement for cloud-computing incorporation.

5.4 Translational Future Scope

The results of this thesis create a solid basis for multiple advanced, PhD-level research initiatives focused on transforming preclinical drug safety.

- Validation of Metabolic Flux (In-Vitro): The computational results produced by this research necessitate experimental validation in the laboratory. Future efforts will shift from in-silico structural interactions to in-vitro metabolic assessments. Human cell lines will be subjected to different concentrations of DPH to measure real-time cellular lipid deprivation (FASN inhibition) and one-carbon/folate pool reduction (DHFR/FOLH1 inhibition) through mass spectrometry-based metabolomics.
- Predictive Toxicology and Machine Learning: The combination of assessed off-target binding affinities with tissue-specific transcriptomic information from the Human Protein Atlas offers a chance to develop deep-learning models. These models might reliably forecast organ-specific toxicity risks (e.g., cardiotoxicity, hepatotoxicity) for any small molecule prior to its entry into clinical trials.
- Expansion of the Polypharmacological Framework: The verified modular pipeline created in this research will be broadened to methodically evaluate additional frequently used, over-the-counter (OTC) small-molecule medications. Evaluating these apparently "safe" compounds using the AlphaFold-enhanced human proteome will contribute to building a thorough database of concealed polypharmacological networks, eventually aiding in safer drug repurposing and development.

5.4.1 Pharmacogenomics and Personalized Safety Profiling

An important future direction for this study is the incorporation of pharmacogenomics to account for individual differences in drug responses. Although our research highlights GABRG2 as a significant off-target for Diphenhydramine, the clinical occurrence of sedation varies among individuals. This "phenotypic variability" might originate from genetic polymorphisms present in the off-target genes themselves.

Particular Single Nucleotide Polymorphisms (SNPs) within the GABRG2 gene (like rs211037) have been linked to changes in GABA-A receptor sensitivity and expression in the CNS. People with certain allelic variants may have a binding pocket that is more "receptive" to

Diphenhydramine, resulting in the intense drowsiness or "hangover" effect felt by some users but not by others. Integrating our proteome-wide docking scores with high-throughput sequencing data (such as that from the 1000 Genomes Project) enables us to progress toward Personalized Safety Profiling.

In this approach, a patient's genetic profile would be evaluated against the "off-target interactome" of a drug such as DPH. A patient with a high-affinity structural variant of an off-target such as GABRG2 or DHFR would be categorized as "high-risk" for certain adverse reactions. This shift from a "one-size-fits-all" safety approach to a genome-informed toxicity prediction model signifies the upcoming challenge in reducing ADRs for versatile, first-generation drugs.

REFERENCES

1. Berman, H. M., Westbrook, J., Feng, Z., Gilliland, G., Bhat, T. N., Weissig, H., ... & Bourne, P. E. (2000). The Protein Data Bank. Nucleic Acids Research, 28(1), 235-242.
2. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., ... & Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596(7873), 583-589.
3. Kanehisa, M., & Goto, S. (2000). KEGG: Kyoto encyclopedia of genes and genomes. Nucleic Acids Research, 28(1), 27-30.
4. Liu, Y., Grimm, M., Dai, W. T., Hou, M. C., Xiao, Z. X., & Cao, Y. (2020). CB-Dock: a web server for cavity detection-guided protein–ligand blind docking. Acta Pharmacologica Sinica, 41(1), 138-144.
5. Mendez, D., Gaulton, A., Bento, A. P., Chambers, J., De Veij, M., Félix, E., ... & Leach, A. R. (2019). ChEMBL: towards direct deposition of bioassay data. Nucleic Acids Research, 47(D1), D930-D940.
6. Rao, M. S., Gupta, R., et al. (2019). Polypharmacology and computational discovery of off-targets. Frontiers in Big Data, 2, 25.
7. Reulecke, I., Lange, G., Albrecht, J., Klein, R., & Rarey, M. (2008). Towards an integrated description of hydrogen bonding and dehydration: decreasing false positives in virtual screening with the HYDE scoring function. ChemMedChem, 3(6), 885-897.
8. Sliwoski, G., Kothiwale, S., Meiler, J., & Lowe, E. W. (2014). Computational methods in drug discovery. Pharmacological Reviews, 66(1), 334-395.
9. Szklarczyk, D., Gable, A. L., Nastou, K. C., Lyon, D., Kirsch, R., Pyysalo, S., ... & von Mering, C. (2021). The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets. Nucleic Acids Research, 49(D1), D605-D612.
10. Trott, O., & Olson, A. J. (2010). AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. Journal of Computational Chemistry, 31(2), 455-461.

11. Uhlén, M., Fagerberg, L., Hallström, B. M., Lindskog, C., Oksvold, P., Mardinoglu, A., ... & Pontén, F. (2015). Proteomics. Tissue-based map of the human proteome. Science, 347(6220), 1260419.
12. UniProt Consortium. (2023). UniProt: the universal protein knowledgebase in 2023. Nucleic Acids Research, 51(D1), D523-D531.
13. Volkamer, A., Kuhn, D., Grombacher, T., Rippmann, F., & Rarey, M. (2012). Combining geometric and pharmacophore-based derisking of protein binding sites with DoGSiteScorer. Journal of Chemical Information and Modeling, 52(2), 360-372.
14. Wishart, D. S., Feunang, Y. D., Guo, A. C., Lo, E. J., Marcu, A., Grant, J. R., ... & Wilson, M. (2018). DrugBank 5.0: a major update to the DrugBank database for 2018. Nucleic Acids Research, 46(D1), D1074-D1082.
15. Krivák, R., & Hoksza, D. (2018). P2Rank: machine learning based tool for rapid and accurate prediction of ligand binding sites from 3D protein structures. *Journal of Cheminformatics*, 10(1), 39.
16. van Kempen, M., et al. (2023). Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, 41(5), 673-682.
17. O'Boyle, N. M., et al. (2011). Open Babel: An open chemical toolbox. *Journal of Cheminformatics*, 3(1), 33.
18. Masiulis, S., et al. (2019). GABAA receptor signalling mechanisms revealed by structural pharmacology. *Nature*, 565(7740), 454-459.
19. Yan, R., et al. (2020). Structural basis for the recognition of SARS-CoV-2 by full-length human ACE2. *Science*, 367(6485), 1444-1448.
20. Shimamura, T., Shiroishi, M., Weyand, S., Tsujimoto, H., Winter, G., Katritch, V., ... & Iwata, S. (2011). Structure of the human histamine H1 receptor complex with doxepin. *Nature*, 475(7354), 65-70.
21. Tang, S. S., Chen, R. Z., Lin, W., Lin, Q., Zhu, W., & Xu, X. (2022). Vina-GPU: A Heterogeneous Computing-Accelerated AutoDock Vina. *Journal of Chemical Information and Modeling*, 62(14), 3420-3428.